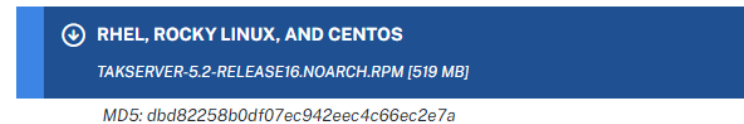


TAK Server Deployment – Command Line

<https://mytecknet.com/lets-build-a-tak-server/>

TAK Server Deployment – Command Line

- If you haven't already, download TAK-SERVER
 - Go to the following link and download Rocky8 release version of TAK Server on your local machine to the “TAK Server Install” folder
 - <https://tak.gov/products/tak-server>



SSH into Virtual Machine

- Open your choice of terminal (Default: PowerShell) or you can use PuTTY.
 - ssh into your VM by running the following command:
 - **ssh <username>@<IP address>**
 - You should be prompted asking if you are sure you want to connect, which you will respond “yes” and hit enter.
 - It will then ask you for your password.

```
PS C:\Users\huntr> ssh takadmin@192.168.1.101
The authenticity of host '192.168.1.101 (192.168.1.101)' can't be established.
ECDSA key fingerprint is SHA256:BbfKUAQONLcOIoiigQQK0/VCbcWznbdIN0FZY8u/KOQ.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.101' (ECDSA) to the list of known hosts.
takadmin@192.168.1.101's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Fri Aug  2 08:26:39 2024
```

Prerequisites

- Increase JVM threads
 - **echo -e "* soft nofile 32768\n* hard nofile 32768\n" | sudo tee --append /etc/security/limits.conf**
 - Each 'tab' is six spaces when typing manually
 - Tail to view the end contents of a file
 - **sudo tail -n 15 /etc/security/limits.conf**
- Install nano or vim
 - **sudo dnf install nano -y**
 - **sudo dnf install vim -y**
- Install packages for Enterprise Linux (EPEL)
 - Enable Powertools
 - **sudo dnf config-manager --set-enabled powertools**
 - Rocky Linux 9
 - **sudo dnf config-manager --set-enabled crb**
 - Install EPEL Rocky linux 8
 - **sudo dnf install epel-release -y**

```
[takadmin@localhost tak]$ sudo tail -n 15 /etc/security/limits.conf
#<domain>      <type>  <item>          <value>
#
#*              soft    core            0
#*              hard    rss             10000
#@student       hard    nproc           20
#@faculty       soft    nproc           20
#@faculty       hard    nproc           50
#ftp            hard    nproc           0
#@student       -       maxlogins       4

# End of file
* soft nofile 32768
* hard nofile 32768
[takadmin@localhost tak]$
```

```
[takadmin@localhost ~]$ sudo dnf install nano -y

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for takadmin:
Rocky Linux 8 - AppStream                2.4 MB/s | 11 MB   00:04
Rocky Linux 8 - BaseOS                  1.7 MB/s | 5.0 MB   00:02
Rocky Linux 8 - Extras                   12 kB/s | 14 kB   00:01
Package nano-2.9.8-1.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

Prerequisites

- Install PostgreSQL/PostGIS 15
 - **sudo rpm --import**
<https://download.postgresql.org/pub/repos/yum/keys/PGDG-RPM-GPG-KEY-RHEL>
 - **sudo dnf install**
https://download.postgresql.org/pub/repos/yum/reporepms/EL-8-x86_64/pgdg-redhat-repo-latest.noarch.rpm -y
 - **sudo dnf -qy module disable postgresql**

```
[t@kadmin@localhost ~]$ # Install PostgreSQL/PostGIS 15 for Rocky Linux/RHEL 8
[t@kadmin@localhost ~]$ sudo rpm --import https://download.postgresql.org/pub/repos/yum/keys/PGDG-RPM-GPG-KEY-RHEL
[t@kadmin@localhost ~]$ sudo dnf install https://download.postgresql.org/pub/repos/yum/reposrums/EL-8-x86_64/pgdg-redhat-repo-latest.noarch.rpm -y
Extra Packages for Enterprise Linux 8 - x86_64                               3.5 MB/s | 14 MB    00:03
Last metadata expiration check: 0:00:08 ago on Fri 02 Aug 2024 08:43:26 AM EDT.
pgdg-redhat-repo-latest.noarch.rpm                                         18 kB/s | 15 kB    00:00
Dependencies resolved.
===== Package
Architecture      Version      Repository      Size
=====
pgdg-redhat-repo  noarch      42.0-43PGDG     @commandline    15 k
=====Installing:
Transaction Summary
=====Install 1 Package
Total size: 15 k
Installed size: 15 k
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing          :
Installing         : pgdg-redhat-repo-42.0-43PGDG.noarch      1/1
Verifying          : pgdg-redhat-repo-42.0-43PGDG.noarch      1/1
Installed:
pgdg-redhat-repo-42.0-43PGDG.noarch
Complete!
```

```
[takadmin@localhost ~]$ sudo dnf -qy module disable postgresql
Importing GPG key 0x08B40D20:
  Userid : "PostgreSQL RPM Repository <pgsql-pkg-yum@lists.postgresql.org>"
  Fingerprint: D4BF 08AE 67A0 B4C7 A1DB CCD2 40BC A2B4 08B4 0D20
  From    : /etc/pki/rpm-gpg/PGDG-RPM-GPG-KEY-RHEL
Importing GPG key 0x08B40D20:
  Userid : "PostgreSQL RPM Repository <pgsql-pkg-yum@lists.postgresql.org>"
  Fingerprint: D4BF 08AE 67A0 B4C7 A1DB CCD2 40BC A2B4 08B4 0D20
  From    : /etc/pki/rpm-gpg/PGDG-RPM-GPG-KEY-RHEL
Importing GPG key 0x08B40D20:
  Userid : "PostgreSQL RPM Repository <pgsql-pkg-yum@lists.postgresql.org>"
  Fingerprint: D4BF 08AE 67A0 B4C7 A1DB CCD2 40BC A2B4 08B4 0D20
  From    : /etc/pki/rpm-gpg/PGDG-RPM-GPG-KEY-RHEL
Importing GPG key 0x08B40D20:
  Userid : "PostgreSQL RPM Repository <pgsql-pkg-yum@lists.postgresql.org>"
  Fingerprint: D4BF 08AE 67A0 B4C7 A1DB CCD2 40BC A2B4 08B4 0D20
  From    : /etc/pki/rpm-gpg/PGDG-RPM-GPG-KEY-RHEL
Importing GPG key 0x08B40D20:
  Userid : "PostgreSQL RPM Repository <pgsql-pkg-yum@lists.postgresql.org>"
  Fingerprint: D4BF 08AE 67A0 B4C7 A1DB CCD2 40BC A2B4 08B4 0D20
  From    : /etc/pki/rpm-gpg/PGDG-RPM-GPG-KEY-RHEL
```

Prerequisites

- Update Rocky Linux
 - **sudo dnf update -y**
- Install Java OpenJDK
 - **sudo dnf install java-17-openjdk-devel -y**

```
Installed:
copy-jdk-configs-4.0-2.el8.noarch
java-17-openjdk-devel-1:17.0.12.0.7-2.el8.x86_64
javapackages-filesystem-5.3.0-2.module+el8.3.0+125+5da1ae29.noarch
ttmkfdir-3.0.9-54.el8.x86_64
xorg-x11-fonts-Type1-7.5-19.el8.noarch

java-17-openjdk-1:17.0.12.0.7-2.el8.x86_64
java-17-openjdk-headless-1:17.0.12.0.7-2.el8.x86_64
lksctp-tools-1.0.18-3.el8.x86_64
tzdata-java-2024a-1.el8.noarch

Complete!
[takadmin@localhost ~]$
```

Installation Instructions

- Copy files from host machine to your VM
 - Open another terminal in the specific folder that holds the TAK Server download.
 - Navigate to the file with File Explorer.
 - At the top of File Explorer select “File” -> “open with PowerShell”
 - **scp takserver-5.*.rpm <username>@<IP address>:~/**
 - If needed, you can specify the entire name of the file instead of using “.*.”
 - **scp takserver-5.2-RELEASE16.noarch.rpm <username>@<IP address>:~/**
 - You will be prompted for the password and the file transfer will start.

```
PS C:\Users\huntr\Downloads> scp takserver-5.2-RELEASE16.noarch.rpm takadmin@192.168.1.101:~/
takadmin@192.168.1.101's password:
takserver-5.2-RELEASE16.noarch.rpm                                100% 519MB 25.4MB/s 00:20
PS C:\Users\huntr\Downloads> _
```

Installation Instructions

- In our ssh session window to the VM you should be able to now see the file we transferred by running:
 - **ls**
- Install the TAK Server file we just transferred.
 - **sudo dnf install takserver-5.*.rpm -y**
 - If you need to specify the server's name you can, but since we only have one file named this, we can use the abbreviation.
- If needed, enter your password and the download should begin.

Installation Instructions

- Check SELinux status (Security-Enhanced Linux)
 - **getenforce**
 - If SELinux is enforcing, we need to apply the SELinux profile for the TAK Server by installing an extra package before applying the SELinux profile.
 - Install checkpolicy
 - **sudo dnf install checkpolicy**
 - enter your password if needed
 - Apply the TAK Server SELinux profile
 - **cd /opt/tak && sudo ./apply-selinux.sh**

```
[takadmin@localhost ~]$ sudo dnf install checkpolicy
[sudo] password for takadmin:
Last metadata expiration check: 0:42:10 ago on Fri 02 Aug 2024 08:44:09 AM EDT.
Package checkpolicy-2.9-1.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[takadmin@localhost ~]$
```

```
[takadmin@localhost tak]$ cd /opt/tak && sudo ./apply-selinux.sh
SELinux Enabled. Applying policy.
Enforcing policy
[takadmin@localhost tak]$
```

Initial Configuration

- Switch to the “tak” user
 - **sudo su tak**
- Change the directory
 - **cd /opt/tak/certs**
- Open the file
 - **vi cert-metadata.sh**
 - To edit the file press “i”
 - Change the following fields: (additional instructions are at the top)
 - COUNTRY=<enter country>
 - STATE=<enter state>
 - CITY=<enter city>
 - ORGANIZATION=<enter org>
 - ORGANIZATION UNIT=<enter org unit>
 - Optionally you can modify the password for the cert located at “atakatak” leaving the “-” before it
 - To exit:
 - Press the “esc” key
 - Type the following: :wq!

```
# Common configuration for all certificates
# Edit these fields to be appropriate for your organization
# If they are left blank, they will not be included. Do not leave COUNTRY
# blank (you may set it to "XX" if you want to be obtuse).
#
# Values for each may be optionally set as environment variables.
# Replace variables such as ${STATE} and ${CITY} as needed.
#

COUNTRY=US
STATE=${STATE}
CITY=${CITY}
ORGANIZATION=${ORGANIZATION:-TAK}
ORGANIZATIONAL_UNIT=${ORGANIZATIONAL_UNIT}

CAPASS=${CAPASS:-atakatak}
PASS=${PASS:-$CAPASS}

## subdirectory to put all the actual certs and keys in
DIR=files

##### don't edit below this line #####
```

Initial Configuration

- Creating the Root Certificate Authority

- **./makeRootCa.sh** or **./makeRootCa.sh --ca-name <CommonName>**

- Example: **./makeRootCa.sh --ca-name TAK-ROOT-CA-01**

- Creating the Intermediate Certificate Authority

- **./makeCert.sh ca** or **./makeCert.sh ca <CommonName>**

- Example: **./makeCert.sh ca TAK-ID-CA-01**

- Once you hit enter it'll come to a point where it'll
me to move files around"

- type: y
- hit enter

```
bash-4.4$ ./makeRootCa.sh --ca-name TAK-Server-CDL
Making a CA for /C=US/ST=CA/L=CA/O=TEST/OU=TESTING/CN=TAK-Server-CDL
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'ca-do-not-share.key'
-----
Certificate was added to keystore
Using configuration from ../config.cfg
bash-4.4$
```

```
bash-4.4$ ./makeRootCa.sh --ca-name TAK-Server-CDL
Making a CA for /C=US/ST=CA/L=CA/O=TEST/OU=TESTING/CN=TAK-Server-CDL
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'ca-do-not-share.key'
-----
Certificate was added to keystore
Using configuration from ../config.cfg
bash-4.4$ ./makeCert.sh ca TAK-Server-CDL
ca
TAK-Server-CDL
Making a ca cert for /C=US/ST=CA/L=CA/O=TEST/OU=TESTING/CN=TAK-Server-CDL
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'TAK-Server-CDL.key'
-----
Signature ok
subject=C = US, ST = CA, L = CA, O = TEST, OU = TESTING, CN = TAK-Server-CDL
Getting CA Private Key
Certificate was added to keystore
Importing keystore TAK-Server-CDL-signing.p12 to TAK-Server-CDL-signing.jks...
Using configuration from ../config.cfg
Do you want me to move files around so that future server and client certificates are signed by this new CA? [y/n]
y
bash-4.4$
```

Initial Configuration

- Creating the Server Certificate
 - **`./makeCert.sh server takserver`**
 - By default, the name is takserver, if you wish to use a different name, we will need to update our files.
 - **`sed -i 's/takserver/<TAKServerName>/g' /opt/tak/CoreConfig.example.xml`**

```
bash-4.4$ ./makeCert.sh server TAK-Server-CDL
server
TAK-Server-CDL
Making a server cert for /C=US/ST=CA/L=CA/O=TEST/OU=TESTING/CN=TAK-Server-CDL
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'TAK-Server-CDL.key'
-----
Signature ok
subject=C = US, ST = CA, L = CA, O = TEST, OU = TESTING, CN = TAK-Server-CDL
Getting CA Private Key
Importing keystore TAK-Server-CDL.p12 to TAK-Server-CDL.jks...
bash-4.4$ sed -i 's/takserver/TAK-Server-CDL/g' /opt/tak/CoreConfig.example.xml
bash-4.4$
```

Initial Configuration

- Modifying the Core Configuration
 - Modify the default truststore-root with out new truststore-<CommonName>
 - **sed -i 's/truststore-root/truststore-<CACommonName>/g' /opt/tak/CoreConfig.example.xml**
 - Example: **sed -i 's/truststore-root/truststore-TAK-ID-CA-01/g' /opt/tak/CoreConfig.example.xml**
 - You can verify the changes by running to following command:
 - **cat /opt/tak/CoreConfig.example.xml | grep truststore-**
 - If you changed the password for the certificates, you would also need to update that within the CoreConfig.example.xml file.
 - **sed -i 's/atakatak/<password>/g' /opt/tak/CoreConfig.example.xml**

```
bash-4.4$ sed -i 's/takserver/TAK-Server-CDL/g' /opt/tak/CoreConfig.example.xml
bash-4.4$ sed -i 's/truststore-root/truststore-TAK-Server-CDL/g' /opt/tak/CoreConfig.example.xml
bash-4.4$ # Verify our changes
bash-4.4$ cat /opt/tak/CoreConfig.example.xml | grep truststore-
truststore="JKS" truststoreFile="certs/files/truststore-TAK-Server-CDL.jks" truststorePass="atakatak">
bash-4.4$
```

Initial Configuration - Starting the TAK Server

- Exit out of the tak user session
 - **exit**
- Enable the takserver.service, allowing it to start at system startup
 - **sudo systemctl enable takserver.service**
- Start the TAK Server service
 - **sudo systemctl start takserver.service**

```
bash-4.4$ exit
exit
[takadmin@localhost tak]$ sudo systemctl enable takserver.service
[sudo] password for takadmin:
takserver.service is not a native service, redirecting to systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable takserver
[takadmin@localhost tak]$ sudo systemctl start takserver.service
[takadmin@localhost tak]$
```

Initial Configuration - Starting the TAK Server

- Verify the server is running:
 - Monitor the `/opt/tak/logs/` directory for the `takserver-messaging.log`
 - **`ls -l /opt/tak/logs/`**
 - make sure the `takserver-messaging.log` file was created.
 - Tail a streaming output of the end of the log
 - **`tail -f /opt/tak/logs/takserver-messaging.log`**
 - INFO c.b.marti.nio.netty.NioNettyBuilder - Successfully Started Netty Server for TlsServerInitializer on Port 8089 with watermark WriteBufferWaterMark(low: 2048, high: 4096)
 - This lets us know that the client-to-server listener is started and is accepting clients
 - INFO com.bbn.marti.nio.server.NioServer - Server started
 - This tells us the services have started
 - INFO com.bbn.marti.util.VersionBean - TAK Server version 5.0-RELEASE-29-HEAD
 - This lets me know what version is running and that all services have started

Initial Configuration - Creating the Admin Certificate

- Switch to the “tak” user
 - **sudo su tak**
- Change the directory to certs
 - **cd /opt/tak/certs**
- Make the Admin Certificate
 - **./makeCert.sh client <CommonName>**
 - Example: **./makeCert.sh client webadmin**
- Modify the certificate to provide admin access
 - **java -jar /opt/tak/utils/UserManager.jar certmod -A /opt/tak/certs/files/<commonName>.pem**
 - Example: **java -jar /opt/tak/utils/UserManager.jar certmod -A /opt/tak/certs/files/webadmin.pem**

```
bash-4.4$ java -jar /opt/tak/utils/UserManager.jar certmod -A /opt/tak/certs/files/serveradmin.pem
New User Added:
  Username:      'serveradmin'
  Role:          ROLE_ADMIN
  Fingerprint:   FE:BF:92:1C:7C:84:37:17:BB:EC:7C:AB:3D:A6:36:34:EB:BD:67:5E:A0:7F:83:C2:07:BF:C7:5C:DC:B2:3E:96
  Groups (read and write permission):
    __ANON__
bash-4.4$
```


Initial Configuration

- Exit the tak user session
 - **exit**
- Change the directory to the Linux home directory
 - **cd ~**
- Copy the admin cert to our home directory
 - **sudo cp -v /opt/tak/certs/files/webadmin.p12 .**
- Copy the truststore cert to our home directory
 - **sudo cp -v /opt/tak/certs/files/truststore-TAK-ID-CA-01.p12 .**
- Change the ownership to our Linux user
 - **sudo chown -R <username>:<usergroup> /home/<username>**
 - Example: **sudo chown -R takadmin:takadmin /home/takadmin**
 - If you don't do this, you can't copy the files from the server to your development machine.
 - You can use "ls" to verify that the file is in the right place .

Configuring the Firewall

- Determine the active firewall
 - **sudo firewall-cmd --get-active-zones**
 - the default is public in most installations
- Modify the zone to allow secure client-to-server communications
 - **sudo firewall-cmd --zone=public --add-port 8089/tcp --add-port 8443/tcp --add-port 8446/tcp --permanent**
- Reload the firewall to apply changes
 - **sudo firewall-cmd --reload**
- Verify the changes to the firewall
 - **sudo firewall-cmd --list-all**

SERVICE	PROTOCOL	PORT	SOURCE	DESTINATION	DIRECTION
TAK Signaling	TCP/S	8089	Client	Server	IN
TAK API WebUI WebTAK	TCP/S	8443	Client	Server	IN
Federation	TCP/S	8444(Legacy) 9000(v1) 9001(v2)	Server	Server	IN
Certificate Provisioning	TCP/S	8446	Client	Server	IN

```
[takadmin@localhost ~]$ sudo firewall-cmd --get-active-zone
libvirt
  interfaces: virbr0
public
  interfaces: enp0s3 enp0s8
[takadmin@localhost ~]$ sudo firewall-cmd --zone=public --add-port 8089/tcp --add-port 8443/tcp --permanent
success
[takadmin@localhost ~]$ sudo firewall-cmd --reload
success
[takadmin@localhost ~]$ sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3 enp0s8
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 8089/tcp 8443/tcp
  protocols:
  forward: no
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[takadmin@localhost ~]$ sudo chown -R takadmin:takadmin /home/takadmin
[takadmin@localhost ~]$
```

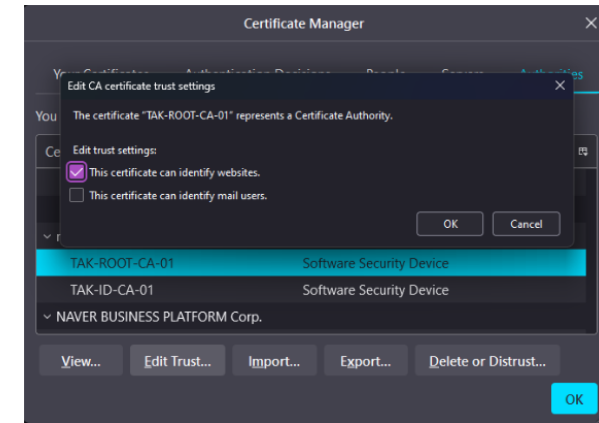
The Administrative Dashboard - Downloading the Admin Cert

- Copy admin file from our Linux server to our local machine.
 - From our local machine open File Explorer to the location, you would like the admin cert file to download to.
 - Once there open the command line to this file:
 - In File Explorer in the top left of the screen selection File -> Open in PowerShell
 - In Powershell type the following:
 - **scp <username>@<IP address>:~/<CommonName>.p12 .**
 - Example: **scp takadmin@192.168.1.104:~/webadmin.p12 .**
 - Don't forget the period at the end to designate the current location to be where the file is downloaded to.
 - Enter the user password when asked for it.

```
1\MASTER_SERVER_NAME_FOLDER_STRUCTURE\Cert Files> scp takadmin@192.168.1.104:~/webadmin.p12 .
takadmin@192.168.1.104's password:
webadmin.p12 100% 4390 1.4MB/s 00:00
PS C:\Users\huntr\Documents\TAK Server\TAK Servers\TAK-Server-CL\MASTER_SERVER_NAME_FOLDER_STRUCTURE-20240731T121233Z-00
1\MASTER_SERVER_NAME_FOLDER_STRUCTURE\Cert Files>
```

The Administrative Dashboard - Firefox

- Open Firefox settings and search for “cert” to find the “view certificates” option.
 - In the certificate manager panel, you want to make sure you select the “Your Certificates” tab and select import.
 - Navigate to where you stored the certificate file and select it.
 - If you are at the folder location for the certificate and can't, see it, change the filtering settings in the bottom right of the File Explorer.
 - It should ask you for the certificate password, if you did not change it the password is: atakatak
 - Once imported select the “Authorities” tab in the “Certificate Manager” panel.
 - Scroll down to find your certificates.
 - Note it will be under the name you gave for the “ORGANIZATION” you entered.
 - Once you found both the certificates one at a time select them and select the “Edit Trust...” button towards the bottom of the panel.
 - Enable the “This certificate can identify websites” option for both.



The Administrative Dashboard - Firefox

- You can close out of the “Certificate Manager” and open a new tab.
 - In the new tab type in the following link:
 - **`https://<IP address>:8443`**
 - A warning risk will pop up, which you will scroll down and select “Accept the Risk and Continue”
 - If done right, you should now see the server dashboard.

